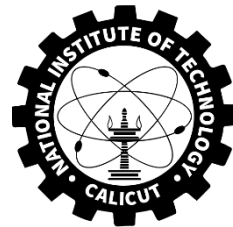


# **Quality, Reliability & Maintenance ME3327E**



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# Weibull Distribution

A Weibull random variable is typically used in reliability analysis to describe the time to failure of mechanical and electrical components. It is a three-parameter distribution. A *Weibull probability density function* is given by

$$f(x) = \frac{\beta}{\alpha} \left( \frac{x - \gamma}{\alpha} \right)^{\beta-1} \exp \left[ - \left( \frac{x - \gamma}{\alpha} \right)^{\beta} \right], x \geq \gamma$$

The parameters are a **location parameter**  $\gamma$  ( $-\infty < \gamma < \infty$ ), a **scale parameter**  $\alpha$  ( $\alpha > 0$ ), and a **shape parameter**  $\beta$  ( $\beta > 0$ ).

The mean and the variance of the Weibull distribution are

$$\mu = E(X) = \gamma + \alpha \Gamma \left( \frac{1}{\beta} + 1 \right)$$

$$\sigma^2 = Var(X) = \alpha^2 \left\{ \Gamma \left( \frac{2}{\beta} + 1 \right) - \left[ \Gamma \left( \frac{1}{\beta} + 1 \right) \right]^2 \right\}$$

# Weibull Distribution

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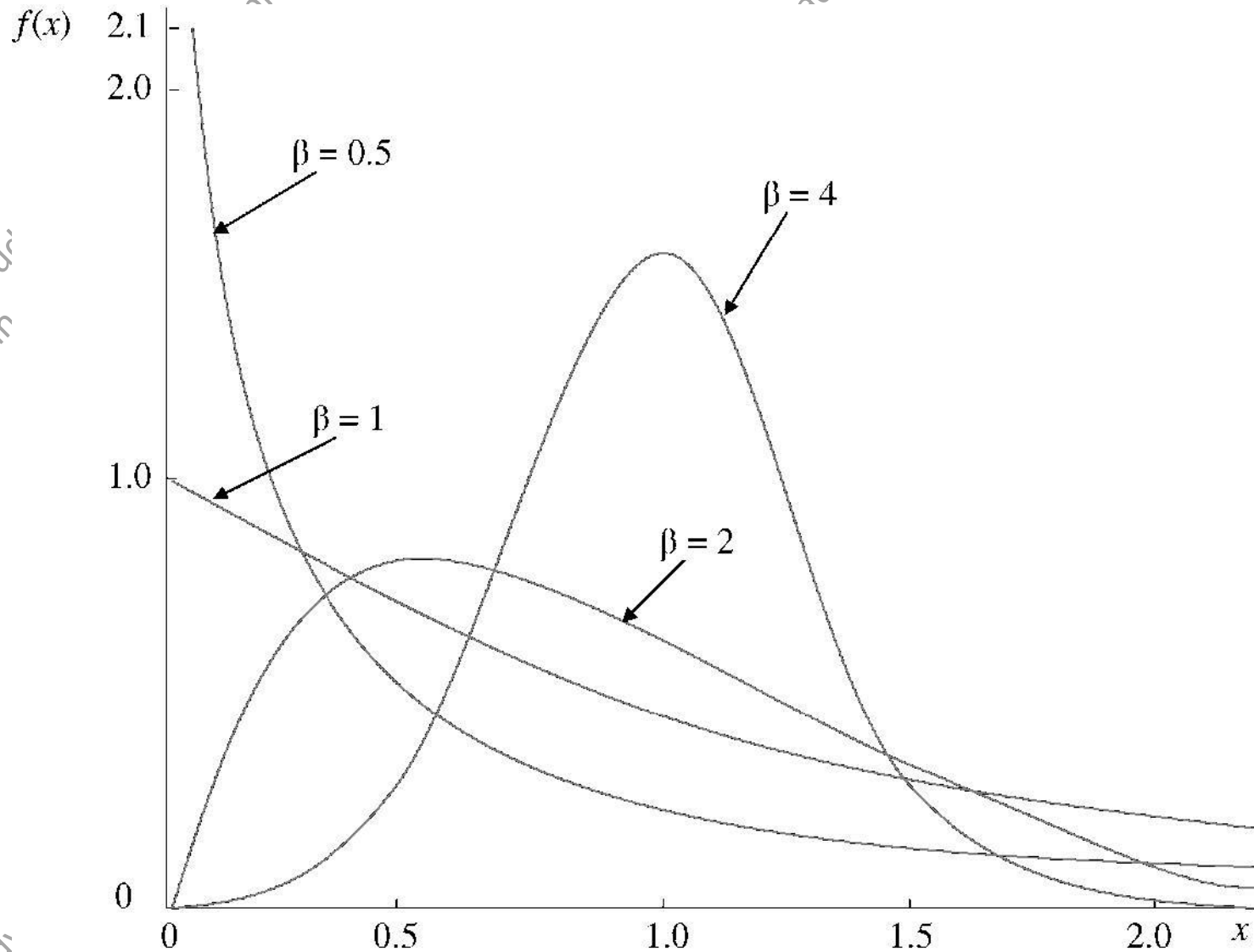
Gamma function,  $\Gamma(t) = \int_0^\infty e^{-x} x^{t-1} dx$

If  $u$  is integer then  $\Gamma(u) = (u - 1)!$

The cumulative distribution function of a Weibull random variable is

given by  $F(x) = 1 - \exp \left[ - \left( \frac{x-\gamma}{\alpha} \right)^\beta \right], x \geq \gamma$

# Weibull Distribution



Weibull probability density functions ( $\gamma=0, \alpha=1, \beta=0.5,1,2,4$ )

# Exercise

The time to failure for a cathode ray tube can be modeled using a Weibull distribution with parameters  $\gamma = 0, \beta = \frac{1}{3}$  and  $\alpha = 200 \text{ hours}$ .

- Find the mean time to failure and its standard deviation.
- What is the probability of a tube operating for at least 800 hours?

| Feature      | Normal Distribution                    | Exponential Distribution                        | Weibull Distribution                                                                   |
|--------------|----------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------|
| Shape        | Symmetric, bell-shaped                 | Right-skewed, steep decline                     | Flexible (skewed, exponential, or bell-shaped)                                         |
| Domain       | $(-\infty, \infty)$                    | $[0, \infty)$                                   | $[0, \infty)$                                                                          |
| Key Property | Driven by Central Limit Theorem        | Memoryless (past history doesn't affect future) | Flexible hazard rate (models age/wear)                                                 |
| Parameters   | Mean ( $\mu$ ), Std Dev ( $\sigma$ )   | Rate ( $\lambda$ )                              | Shape ( $\beta$ ), Scale ( $\alpha$ )                                                  |
| Failure Rate | Increases sharply after mean           | Constant over time                              | Decreasing ( $\beta < 1$ ),<br>Constant ( $\beta = 1$ ),<br>Increasing ( $\beta > 1$ ) |
| Primary Use  | Natural variations, measurement errors | Time between random Poisson events              | Reliability engineering, equipment lifespan                                            |

# Gamma Distribution

Another important distribution with applications in reliability analysis is the gamma distribution. Its probability density function is given by

$$f(x) = \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x}, x \geq 0$$

Where  $k$  is a shape parameter,  $k > 0$ , and  $\lambda$  is a scale parameter,  $\lambda > 0$ . The mean and the variance of the gamma distribution are

$$\mu = \frac{k}{\lambda}, \quad \sigma^2 = \frac{k}{\lambda^2}$$

# Lognormal Distribution

A random variable  $X$  has a lognormal distribution if  $\ln(x)$  has a normal distribution with mean  $\mu$  and variance  $\sigma^2$ , where  $\ln$  represents the natural logarithm. Its probability density function is given by  $f(x) = \frac{1}{\sqrt{2\pi\sigma x}} \exp\left[\frac{-(\ln(x)-\mu)^2}{2\sigma^2}\right], x > 0$

The mean and variance of the lognormal distribution are

$$E(X) = \exp\left(\mu + \frac{\sigma^2}{2}\right), \quad Var(X) = \exp(2\mu + \sigma^2)[\exp(\sigma^2) - 1]$$

The cumulative distribution function is expressed as

$$F(x) = \Phi\left[\frac{\ln(x) - \mu}{\sigma}\right], \quad x > 0$$

Where  $\Phi(\cdot)$  represents the CDF of the standard normal.



# Inferential Statistics

The two main procedures of inferential statistics are estimation (point and interval) and hypothesis testing

## Sampling Distributions

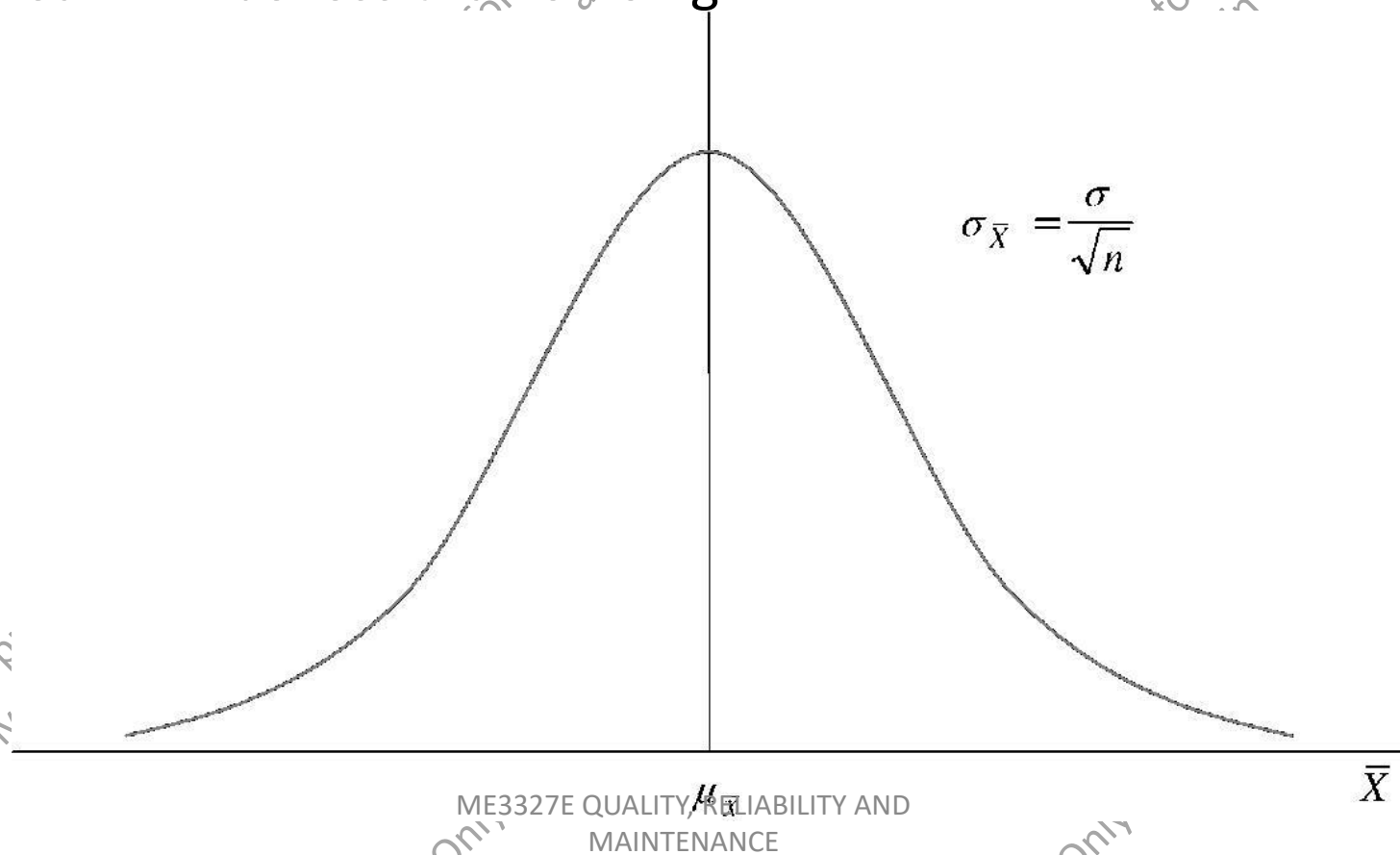
A sample variance is an estimator of the population variance. Studying the behavior of these estimators through repeated sampling allows us to draw conclusions about the corresponding parameters. The behavior of an estimator in repeated sampling is known as the sampling distribution of the estimator, which is expressed as the probability distribution of the statistic.

The sample mean is one of the most widely used estimators in quality control because analysts frequently need to estimate the population mean. It is therefore of interest to know the sampling distribution of the sample mean; this is described by the central limit theorem.

# Sampling Distributions

## Exercise:

The tuft bind strength of a synthetic material used to make carpets is known to have a mean of 50 kg and a standard deviation of 10 kg. If a sample of size 40 is randomly selected, what is the probability that the sample mean will be less than 52.5 kg?



# Estimation of Product and Process Parameters

There are two types of estimation: *point estimation* and *interval estimation*

In *point estimation*, a single numerical value is obtained as an estimate of the population parameter.

In *interval estimation*, a range or interval is determined such that there is some desired level of probability that the true parameter value is contained within it. Interval estimates are also called confidence intervals.

# Confidence Interval for the Mean

## 1. Variance known:

$$\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

## 2. Variance Unknown

$$\bar{X} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

# Confidence Interval for the Difference Between Two Means

1. Variance known:

$$(\bar{X}_1 - \bar{X}_2) - Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{X}_1 - \bar{X}_2) + Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

2. Variance Unknown

$$(\bar{X}_1 - \bar{X}_2) - t_{\alpha/2, v} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{X}_1 - \bar{X}_2) + t_{\alpha/2, v} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$v = \frac{\left( s_1^2/n_1 + s_2^2/n_2 \right)^2}{\left( s_1^2/n_1 \right)^2 / (n_1 - 1) + \left( s_2^2/n_2 \right)^2 / (n_2 - 1)}$$

# Confidence Interval for a Proportion

$$\hat{p} - Z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \leq p \leq (\bar{X}_1 - \bar{X}_2) + Z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

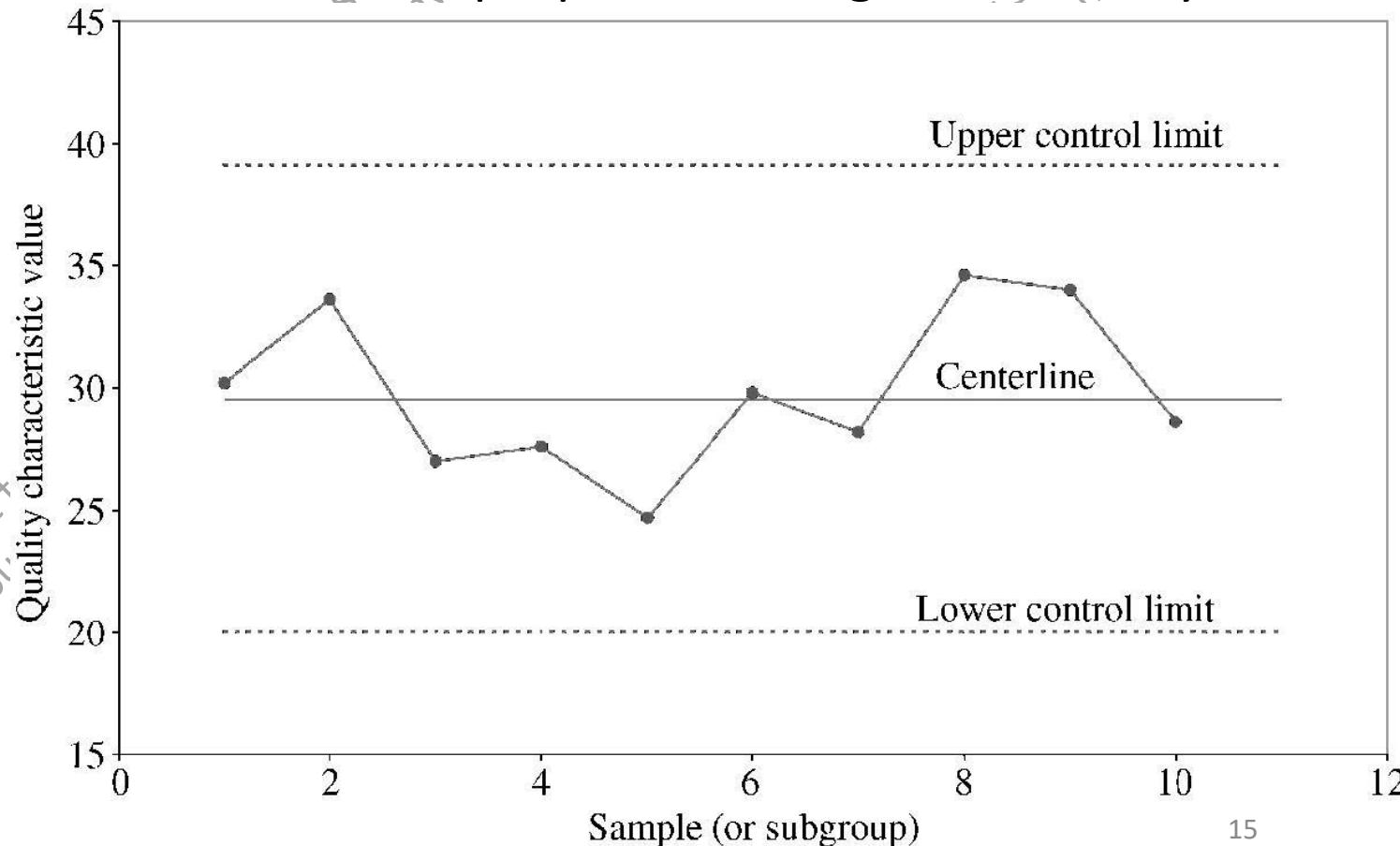
# Confidence Interval for the Difference Between Two Binomial Proportions

$$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \leq p_1 - p_2$$

$$\leq (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

# Statistical Process Control

- A control chart is a graphical tool for monitoring the activity of an ongoing process.
- Control charts are sometimes referred to as Shewhart control charts, because Walter A. Shewhart first proposed their general theory.



# Benefits of Control Charts

- When to take corrective action.
- Type of remedial action necessary.
- When to leave a process alone.
- Process capability.
- Possible means of quality improvement.

## Cause of Variations:

- Special Cause (*As per Deming 15%*): If an observation falls outside the control limits or a non-random pattern is exhibited, special causes are assumed to exist, and the process is said to be out of control.
- Common Cause (*As per Deming 85%*): If quality characteristic values are within control limits and no non-random pattern is visible, it is assumed that a system of common causes exists and that the process is in a state of statistical control.